

16 kHz waveform $x(t)$

+ white noise, $\sigma = \text{RMS}(\text{speech}) \cdot 10^{(\text{dB}/20)}$

envelope (mel) path

STFT 512/256 hann
↓ 16 mel bands ↓ log
↓ affine $(v+10)/10$, clamp ≥ 0

rows $A_{\text{mel}}(t)$ at 62.5 fps
carrier discarded

drive: $\theta_{\text{mel}} \pm g \cdot A_{\text{mel}}(t)$
additive, state-independent

quadrature path

per-band analytic signal
↓ demodulate at band center
↓ baseband phase $\phi_{\text{mel}}(t)$

$(A_{\text{mel}}, \phi_{\text{mel}})$ at 62.5 fps
phase $\in \pm 31$ Hz baseband

drive: $\theta_{\text{mel}} \pm g \cdot A_{\text{mel}} \cdot \sin(\phi_{\text{mel}} - \theta_{\text{mel}})$
Adler, reference-relative

carrier (direct) path

full-clip FFT band-split
16 bands, 96–1,536 Hz
no transcoding, no rate change

rows $x_{\text{mel}}(t)$ at 16 kHz
carrier kept ($\pm$ waveform)

drive: $\theta_{\text{mel}} \pm g \cdot x_{\text{mel}}(t)$
additive, oscillatory

tonotopic broadcast

band r drives every oscillator in grid row r ,
all channels; gain g calibrated per pathway
(units differ 62.5 $\times$); bound: max per-tick kick $< \pi$

what each pathway excites in an untrained field (evidence-tagged)

envelope: state-independent pushes accumulate into readable loudness statistics
(measured: fields hold floor+3 across matrices A–C; weak envelope-rate locking observed, PLV ≈ 0.16)

quadrature: collapsed at every gain tested (measured, 64 $\times$ sweep + matrices) — consistent with
Adler theory: reference-relative torque requires a consistent phase relation, which random
dynamics lack; trainability of that relation is suggested, not shown

carrier: oscillatory push admits injection locking when $|\Delta\omega| \leq A$ (Adler, single-oscillator
idealization; measured: drive-phase PLV rises with gain on frequency-matched rows)

information trade: envelope keeps 0–8 kHz energy at 16 ms resolution; carrier keeps phase, 96–1,536 Hz only